GRAPHS AND MATRICES

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Contents

Abstract. This is a very brief note on **Spectral Graph Theory**.

1. Adjacency Matrices

Given a graph, one can associate various matrices to encode its information. The *adjacency matrix* A of a graph $X = (V, E)$ is the matrix whose rows and columns are indexed by the vertices of X, where $A(x, y)$ equals the number of edges between x and y. When necessary to indicate the dependence on X, we denote A by $A(X)$.

A number λ is called an *eigenvalue* of A if there exists a non-zero vector $u \in \mathbb{R}^n$ such that $Au = \lambda u$. This means that λ is a root of the *characteristic polynomial* $\det(tI_n - A)$.

If n denotes the number of vertices of X, then A is an $n \times n$ real and symmetric matrix, and therefore by linear algebra, it has n real eigenvalues, which we denote as follows: $\lambda_1 \geq \cdots \geq \lambda_n$.

The *spectrum* of X consists of its eigenvalues, including multiplicities.

Theorem 1.1. *Let $X = (V, E)$ be a graph without loops, with adjacency matrix A. For any $x, y \in V$ and any natural number k, the entry $A^k(x, y)$ equals the number of (x, y)-walks of length k.*

Proof. Denote by $w_k(x, y)$ the number of (x, y)-walks of length k. We will show that $A^k(x, y) = w_k(x, y)$ for any $x, y \in V$ and any natural number k using induction on k.

The base case $k = 1$ is clear from the definition of A. Let $k \geq 2$ and assume that $A^{k-1}(x, y) = w_{k-1}(x, y)$ for any $x, y \in V$. Let x and y be two vertices of X. A (x, y)-walk of length k is composed of a (x, z)-walk of length $k - 1$ followed by an edge zy. Hence,

$$w_k(x, y) = \sum_{z : z \sim y} w_{k-1}(x, z).$$

Date: March 2026.

At the same time,

$$A^k(x,y) = \sum_{z \in V} A^{k-1}(x,z)A(z,y) = \sum_{z: z \sim y} A^{k-1}(x,z).$$

From the induction hypothesis, we know that $A^{k-1}(x,z) = w_{k-1}(x,z)$ for each $z \sim y$. Using the above two equations, we get that

$$A^k(x,y) = w_k(x,y).$$

□

Now the above thereom gives the next Corollary 1.1 directly.

Corollary 1.1. *Let $X = (V, E)$ be a graph with adjacency matrix A whose eigenvalues are $\lambda_1 \geq \cdots \geq \lambda_n$. For any natural number k, the number of closed walks of length k in X equals $\sum_{j=1}^{n} \lambda_j^k$.*

Now we shall see a very interesting upper bound on the eigenvalues of a simple graph through the following proposition.

Proposition 1.1. *Let X be a simple graph with n vertices and e edges. If λ is an eigenvalue of the adjacency matrix A of X, then*

$$|\lambda| \leq \sqrt{2e\left(1 - \frac{1}{n}\right)}.$$

Proof. We use the facts that $\mathrm{tr}(A) = 0$ and $\mathrm{tr}(A^2) = 2e$. Since A is real symmetric, all its eigenvalues are real. Let the eigenvalues of A be $\lambda_1, \ldots, \lambda_n$, and let λ be one of them. The desired inequality is equivalent to

$$\lambda^2 \leq \frac{2e(n-1)}{n} \iff \frac{n}{n-1}\lambda^2 \leq \mathrm{tr}(A^2) = \sum_{i=1}^{n} \lambda_i^2.$$

Rewriting, this is equivalent to

$$\sum_{i \neq k} \lambda_i^2 \geq \frac{1}{n-1}\lambda^2,$$

where $\lambda = \lambda_k$. Now, since $\mathrm{tr}(A) = 0$, we have, $\sum_{i \neq k} \lambda_i = -\lambda$. Applying the Cauchy–Schwarz inequality,

$$\sum_{i \neq k} \lambda_i^2 \geq \frac{1}{n-1}\left(\sum_{i \neq k} \lambda_i\right)^2 = \frac{1}{n-1}\lambda^2.$$

This proves the claim. □

2. Bipartite Graphs

A graph $X = (V, E)$ is called *bipartite* if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge of X has one endpoint in V_1 and the other in V_2.

Theorem 2.1. *If X is bipartite, and $\lambda > 0$ is an eigenvalue of the adjacency matrix A of X with multiplicity m, then $-\lambda$ is also an eigenvalue with multiplicity m.*

Proof. Assume that X is a bipartite graph whose partite sets have sizes r and s, respectively. Label the vertices so that the adjacency matrix A has the form

$$A = \begin{pmatrix} 0_r & B \\ B^T & 0_s \end{pmatrix},$$

where B is an $r \times s$ matrix.

Let $\lambda > 0$ be an eigenvalue of A with eigenvector $w = \begin{pmatrix} u \\ v \end{pmatrix}$, where $u \in \mathbb{R}^r$ and $v \in \mathbb{R}^s$, not both zero. Then

$$Aw = \lambda w \implies \begin{cases} \lambda u = Bv, \\ \lambda v = B^T u. \end{cases}$$

Define, $w' = \begin{pmatrix} u \\ -v \end{pmatrix}$.. Then,

$$Aw' = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} u \\ -v \end{pmatrix} = \begin{pmatrix} -Bv \\ B^T u \end{pmatrix} = \begin{pmatrix} -\lambda u \\ \lambda v \end{pmatrix} = (-\lambda) w'.$$

Since $w \neq 0$, we also have $w' \neq 0$. Hence $-\lambda$ is an eigenvalue of A.

A collection of m linearly independent eigenvectors corresponding to λ gives rise to m linearly independent eigenvectors corresponding to $-\lambda$. Thus the multiplicity of $-\lambda$ is at least that of λ. Repeating the argument with $-\lambda$ in place of λ shows that the multiplicities are equal. □

Theorem 2.2 (Spectral Characterization of Bipartite Graphs)**.** *Let X be a graph on n vertices with adjacency matrix A and eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$. The following statements are equivalent:*

(1) *The graph X is bipartite.*
(2) *The spectrum of X is symmetric with respect to zero.*
(3) $t^{-(n \bmod 2)} \det(tI_n - A) \in \mathbb{R}[t^2]$.
(4) *For any odd natural number k, $\sum_{j=1}^{n} \lambda_j^k = 0$.*

Proof. The implication (1) ⇒ (2) was proved in Theorem 2.1. To see that (2) ⇒ (3), note that when n is even, for any positive eigenvalue λ of multiplicity m, we also have the eigenvalue $-\lambda$ with the same multiplicity m. Hence the factors

$$(t-\lambda)^m (t+\lambda)^m = (t^2 - \lambda^2)^m,$$

which is a polynomial in t^2. The eigenvalue 0 has even multiplicity, and therefore $\det(tI_n - A)$ is a polynomial in t^2. When n is odd, a similar argument applies, except that the multiplicity of 0 is odd. Hence $t^{-1}\det(tI_n - A)$ is a polynomial in t^2. The implications (3) ⇒ (2) and (2) ⇒ (4) are trivially true. To see that (4) ⇒ (1), note that condition (4) implies that the number of closed walks of odd length is zero. Therefore, X has no odd cycles, and hence is bipartite. This completes the proof. □

3. Upper Bound on Diameter

The *diameter* of a connected graph $X = (V, E)$ is defined as

$$\operatorname{diam}(X) = \max_{x,y \in V} d(x, y).$$

If A is the adjacency matrix of X, the set of polynomials in A with real coefficients is a vector space over $\mathbb{R}$. It is called the *adjacency algebra* of X, and its dimension equals the number of distinct eigenvalues of X.

Theorem 3.1. *If X is a connected graph whose adjacency matrix has r distinct eigenvalues, then* $\operatorname{diam}(X) \leq r - 1$*.*

Proof. Let $d = \operatorname{diam}(X)$. We show that $I_n, A, \ldots, A^d$ are linearly independent matrices. Since the dimension of the adjacency algebra is r, this implies that $d + 1 \leq r$, which proves the theorem.

To show that $I_n, A, \ldots, A^d$ are linearly independent, we prove that for any $1 \leq k \leq d$, the matrix A^k cannot be written as a linear combination of $I_n, A, \ldots, A^{k-1}$.

Fix $1 \leq k \leq d$ and choose vertices $x, y \in V$ such that $d(x, y) = k$. Then there exists an (x, y)-path of length k, and there are no (x, y)-walks of length ℓ for any $\ell < k$. By Theorem 1.1,

$$A^k(x, y) > 0 \quad \text{and} \quad A^\ell(x, y) = 0 \text{ for all } \ell < k.$$

Therefore, A^k cannot be written as a linear combination of $I_n, A, \ldots, A^{k-1}$. This completes the proof. □

This bound is actually sharp. Take K_n, the adjacency matrix $A = (A(x, y)) = J_n - I_n$, i.e., $A(x, y) = 1 - \delta(x, y)$. We know the characteristic polynomial of J_n is $t^{n-1}(t - n)$, thus the characteristic polynomial of A is $(t + 1)^{n-1}(t - n + 1)$, thus only two distinct eigenvalues of A and diameter of K_n is 1.

4. Laplacian Matrices

First consider X as a directed graph. For each edge e, one of its endpoints becomes the *tail* and the other endpoint is the *head* of the directed edge.

Given such a direction on X, the *directed incidence matrix* N is defined as follows. Its rows are indexed by the vertex set V and its columns by the edge set E. For $x \in V$ and $e \in E$,

$$N(x, e) = \begin{cases} +1, & \text{if } x \text{ is the head of } e, \\ -1, & \text{if } x \text{ is the tail of } e, \\ 0, & \text{otherwise.} \end{cases}$$

Also denote by D the $V \times V$ diagonal matrix whose (x, x)-entry equals the degree of the vertex x. We call D the *diagonal degree matrix* of X.

The *Laplacian matrix* L of a graph X is defined as $L = D - A$, where D is the diagonal degree matrix of X and A is the adjacency matrix of X. This is another important matrix associated to a graph, and we summarize its basic properties below.

We have seen before that spectrum of adjacency matrix can determine whether a graph is bipartite or not but it can not guarantee connectedness of a graph. For example, take C_4 with an isolated vertex and $K_{1,4}$, both have the same spectrum but one is connected and the other is not.

Theorem 4.1. *Let $X = (V, E)$ be a graph and let N be a directed incidence matrix of X.*

1. *The Laplacian matrix L equals NN^T.*
2. *The Laplacian matrix L is real and symmetric, and its eigenvalues are nonnegative.*

(3) *The smallest eigenvalue of L is 0, and its multiplicity equals the number of connected components of X.*

Proof. (1) Let $x, y \in V$. If $x = y$, then the (x,x)-entry of L is $d(x)$. Also,

$$(NN^T)(x,x) = \sum_{e \in E} N(x,e)N^T(e,x) = \sum_{e \in E} N^2(x,e) = d(x).$$

If $x \neq y$ and $x \sim y$, then $L(x,y) = -1$. Let f be the edge connecting x and y. Regardless of its direction, $N(x,f)N(y,f) = -1$, while for any other edge $e \neq f$, $N(x,e)N(y,e) = 0$. Hence,

$$(NN^T)(x,y) = \sum_{e \in E} N(x,e)N(y,e) = -1.$$

If $x \neq y$ and $x \not\sim y$, then $L(x,y) = 0$. In this case, for any $e \in E$, $N(x,e)N(y,e) = 0$, and therefore

$$(NN^T)(x,y) = 0.$$

Thus, $L = NN^T$.

(2) Since D and A are real and symmetric and $L = D - A$, it follows that L is real and symmetric. Let μ be an eigenvalue of L with eigenvector u. Then

$$\mu u = Lu = NN^T u.$$

Multiplying on the left by u^T, we obtain

$$\mu\, u^T u = u^T NN^T u = (N^T u)^T (N^T u).$$

Since $u \neq 0$, we have $u^T u = \sum_{x \in V} u_x^2 > 0$, and hence

$$\mu = \frac{(N^T u)^T (N^T u)}{u^T u} = \frac{\sum_{xy \in E} (u_x - u_y)^2}{\sum_{x \in V} u_x^2} \geq 0.$$

(3) Let $\mathbf{1}$ be the all-ones vector. Then each row of L sums to zero, so $L\mathbf{1} = 0$. Hence 0 is an eigenvalue of L. Assume X is connected. Let u be an eigenvector corresponding to 0. Then

$$0 = \frac{\sum_{xy \in E} (u_x - u_y)^2}{\sum_{x \in V} u_x^2},$$

which implies that $u_x = u_y$ for every edge $xy \in E$. Since X is connected, it follows that $u_x = u_y$ for all $x, y \in V$. Thus u is a scalar multiple of $\mathbf{1}$, and the multiplicity of 0 is 1.

In general, L can be written as a block diagonal matrix whose diagonal blocks are the Laplacian matrices of the connected components of X. The result follows. □

Now we have a sufficient condition for a graph to be connected. If the Laplacian Matrix of the graph has eigenvalue 0 with multiplicity 1, then the graph is connected.

References

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